

1. Introduction. Kanien'kéha (aka Mohawk; Northern Iroquoian) lacks *obligatory* tense morphemes. The temporal interpretation is determined by the choice of aspectual suffixes. To account for morphological differences across the aspectual paradigm, Baker and Travis (1997, 1998) propose that Kanien'kéha has both 'tensed' and 'tenseless' verbal paradigms: more specifically, IMPERFECTIVE and PERFECT aspect forms introduce a time argument, whereas PERFECTIVE aspect verb forms do not.

In this paper, we re-examine this claim and introduce a novel semantic argument regarding counterfactual conditionals, with elicitation data from 4 speakers, showing that Imperfective/Perfect verbs are tensed (argued for independently by Gatchalian 2025), while Perfective verbs are necessarily tenseless. Thus, Kanien'kéha has both *morphologically tenseless* clauses, which lack tense morphology but contain a syntactic/semantic tense (see Matthewson 2006), and *deeply tenseless* clauses, which lack tense entirely (see Pancheva and Zubizarreta 2023). That both options exist in a single language requires that we take seriously the possibility of deep tenselessness, where there is no tense or reference time variable.

2. TAM background. Event verbs in Kanien'kéha obligatorily inflect for one of the three aspects which we label here PERFECTIVE (1a), IMPERFECTIVE (1b) and PERFECT (1c).

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|--------|---------------------|----|-------------------------|----|------------------|
| (1) a. | wa'-k-atshok-o' | b. | k-atshokw-as | c. | wak-atshokw-en |
| | FACT-1SGA-smoke-PFV | | 1SGA-smoke-IPFV | | 1SGP-smoke-PERF |
| | 'I smoked.' | | 'I smoke (habitually).' | | 'I have smoked.' |

The Perfective aspect differs from the other aspects in two ways. First, it *requires* the presence of one of the so-called 'modal prefixes': FACTUAL (1a), FUTURE (2b) or OPTATIVE (2c). The use of a bare perfective form is ruled out (2a). On the other hand, Imperfective and Perfect aspect are *optionally* compatible with some of the modal prefixes, with further derivation (see Baker and Travis 1997, 1998).

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|--------|-------------------|----|--------------------|----|--------------------|
| (2) a. | *k-atshok-o' | b. | en-k-atshok-o' | c. | a-k-atshok-o' |
| | 1SGA-smoke-PFV | | FUT-1SGA-smoke-PFV | | OPT-1SGA-smoke-PFV |
| | Int.: 'I smoked.' | | 'I will smoke.' | | 'for me to smoke' |

Second, Imperfective and Perfect paradigms have an optional overt tense suffix, which is never available on a Perfective aspectual form. Building on this, Gatchalian (2025) argues that Imperfective (1b) and Perfect (1c) forms are covertly tensed. (Details of this argument are omitted here for space.)

Baker and Travis formalize these morphological contrasts as a difference in the presence of a *time argument*: Perfective verbs lack it, while the other aspects obligatorily have it. In essence, this account posits a two-way split in the Kanien'kéha verb paradigm between *tensed* Imperfective and Perfect forms on one hand, and *tenseless* Perfective forms on the other. While the morphological evidence provides grounds for such a split, it does not yet preclude a (covertly) tensed analysis of Perfective verbs. The next section shows that a tensed analysis of Perfective verbs makes incorrect empirical predictions.

3. Novel argument from past counterfactual conditionals. Our argument builds on the interpretive differences of aspectual markers in indicative and counterfactual conditionals. The two constructions are morphologically different: counterfactual conditionals require the use of the OPTATIVE prefix on the verb, while indicative conditional rule out the use of Optative forms. The crucial observation is that in *past counterfactual* conditionals, Perfective forms are infelicitous; this is shown by the contrast in (3). Instead, Optative Perfective verbs get a *future* counterfactual interpretation (data omitted here; cf. Ippolito 2014). In contrast, both the Perfective aspect (4a) and Perfect aspect (4b) can have past reference in *past indicative* conditionals. Thus, it cannot be the Perfective which rules out past reference in (3a).

(3) **Past counterfactual conditional**

Context: Kó:r never does his chores! **He didn't do the dishes and now we have flies in the house.**

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|------|---|------------------------------|------------------------------------|-------|---------|
| a. # | Tóka | a-ha-ks-oharé-nion-' | ne | Kó:r, | |
| | if | OPT-MsgA-dish-wash-DISTR-PFV | NE | Kor | |
| | iah | ki | th-a-ionkwa-nahskw-aién:t-ake' | ne | tsí'ks! |
| | NEG | just | NEG-OPT-1PL.EXP-pet-have.PERF-CONT | NE | fly |
| | Int.: 'If Kó:r had washed the dishes, we wouldn't have flies.' (✗ Opt. Perfective antecedent) | | | | |

- b. Tóka **a-ho-ks-oharé-nion** ne Kó:r,
 if OPT-MSGP-dish-wash-DISTR.PERF NE KOR
 iah ki th-a-ionkwa-nahskw-aién:t-ake' ne tsí'ks!
 NEG just NEG-OPT-1PL.EXP-pet-have.PERF-CONT NE fly
 'If Kó:r had washed the dishes, we wouldn't have flies.' (✓ Opt. Perfect antecedent)

(4) **Past indicative conditional**

Context: Kó:r and Shawátis take turn doing the chores. I know that Kó:r washed the dishes.

- a. Tóka' Kó:r **wa-ha-ks-oharé-nion-**, Shawátis wa'-t-ha-at-ónhew-e'
 if KOR FACT-MSGA-dish-wash-DISTR-PFV Shawatis FACT-DUPL-MSGA-SR-SWEEP-PFV
 'If Kó:r washed the dishes, then Shawatis swept the floor.'
- b. Tóka' Kó:r **ro-ks-oharé-nion**, Shawátis te-ho-tonhé:w-en.
 if KOR MSGP-dish-wash-DISTR.PERF Shawatis DUPL-MSGP-SWEEP-PERF
 'If Kó:r has washed the dishes, then Shawatis has swept the floor.'

Optative forms in counterfactual conditionals are also compatible with the past reference when marked by the Imperfective aspect (5). This shows that (i) Optative on its own does not preclude past tense reference (ii) past tense reference in counterfactuals is not exclusive to the Perfect aspect.

(5) **Optatives are fine in past referring counterfactual conditionals elsewhere**

Context: I introduced Kó:r and Shawátis last year. Shawátis hates smokers. Kó:r was not a smoker, so they became friends

- Tóka' Kó:r **a-ha-tshokwá-hs-eke**, Shawátis a-ho-na'khwá:-s-e'
 if KOR OPT-MSGA-smoke-IPFV-CONT Shawatis OPT-MSGP-be.angry-BEN-PFV
 'If Kó:r were a smoker (at the time), Shawatis would be angry at him.'

To summarize, Perfective aspect is ruled out in past counterfactual sentences (3a); this fact cannot be attributed to the Perfective on its own, which is compatible with past interpretation in general (4a), or the Optative marker, which is also compatible with past contexts elsewhere (3b), (5).

4. Analysis. We argue that the contrast in (3) arises from the *temporal properties* of the aspects. Perfect (and Imperfective) verbs are **tensed** and indexed to a reference time (RT); on the other hand, Perfective verbs **lack tense** and are obligatorily indexed to the evaluation time (EvT).

In counterfactual conditionals, the Optative introduces a modal domain, temporally anchored to some time in the evaluation world (e.g., [Hacquard 2006](#)), and asserts the event holds in worlds in that domain. In *past counterfactuals*, the modal anchor is a past time *t* (denoting RT), thus the domain contains worlds where the antecedent is true, divergently developing from the evaluation world at *t* (cf. *PAST-AS-PAST* theories of counterfactuals, e.g., [Ippolito 2014](#)). Worlds included in this domain are those whose histories match the evaluation world up to the past time *t*, supplied by RT in Perfect/Imperfective aspect verbs. Perfective verbs lack tense and a RT variable altogether, thus the modal anchor is always EvT. That is, the modal domain is always projected from the local *now*, only including worlds which match the history of the evaluation world, diverging from EvT. The divergence of the worlds in the modal domain *from a time prior to EvT* needed to get a past counterfactual is thus unavailable for tenseless Perfective clauses.

In contrast, if Perfective clauses were also tensed, no contrast would be predicted in (3). Recall that Optative only marks the proposition as counterfactual, and does no temporal work. A tensed Perfective would then in principle be able to pick up the past RT and use that as the modal anchor, just like with the Perfect and Imperfective aspects. We conclude that there is no tense mechanism with Perfectives.

5. Discussion. The upshot of this analysis is that Kanien'kéha's tense system is split by aspect. The Imperfective/Perfect paradigm is tensed (often covertly), whereas the Perfective paradigm is *deeply* tenseless. We provided a novel argument that precludes a tensed analysis of Perfective verbs in Kanien'kéha (previously suggested on the basis of morphological facts [Baker and Travis 1997, 1998](#)). The contrast seen in past counterfactual conditionals cannot be captured if both Perfective and Perfect verbs could freely make use of tense to make reference to a past time. Thus, Kanien'kéha is both tensed and tenseless, depending on the aspect we look at; more broadly, languages must allow for the possibility of deeply tenseless clauses as well as superficially tenseless ones.

References

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